

D4 Arithmetic sequences A level only

Name: _____

Class: _____

Date: _____

Time: **128 minutes**

Marks: **107 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

An arithmetic sequence has first term a and common difference d .

The sum of the first 36 terms of the sequence is equal to the square of the sum of the first 6 terms.

(a) Show that $4a + 70d = 4a^2 + 20ad + 25d^2$ (4)

(b) Given that the sixth term of the sequence is 25, find the smallest possible value of a . (5)
(Total 9 marks)

Q2.

(a) Three consecutive terms in an arithmetic sequence are $3e^{-p}$, 5, $3e^p$
 Find the possible values of p . Give your answers in an exact form. (6)

(b) Prove that there is no possible value of q for which $3e^{-q}$, 5, $3e^q$ are consecutive terms of a geometric sequence. (4)
(Total 10 marks)

Q3.

(a) A geometric series begins $420 + 294 + 205.8 + \dots$
 (i) Show that the common ratio of the series is 0.7. (1)

(ii) Find the sum to infinity of the series. (2)

(iii) Write the n th term of the series in the form $p \times q^n$, where p and q are constants. (2)

(b) The first term of an arithmetic series is 240 and the common difference of the series is -8 .

The n th term of the series is u_n .

(i) Write down an expression for u_n . (1)

- (ii) Given that $u_k = 0$, find the value of $\sum_{n=1}^k u_n$.

(4)

(Total 10 marks)

Q4.

An arithmetic series has first term a and common difference d .

The sum of the first 25 terms of the series is 3500.

- (a) Show that $a + 12d = 140$.

(3)

- (b) The fifth term of this series is 100.

Find the value of d and the value of a .

(4)

- (c) The n th term of this series is u_n . Given that

$$33 \left(\sum_{n=1}^{25} u_n - \sum_{n=1}^k u_n \right) = 67 \sum_{n=1}^k u_n$$

find the value of $\sum_{n=1}^k u_n$.

(3)

(Total 10 marks)

Q5.

The arithmetic series

$$23 + 32 + 41 + 50 + \dots + 2534$$

has 280 terms.

- (a) Write down the common difference of the series.

(1)

- (b) Find the 100th term of the series.

(2)

- (c) Find the sum of the 280 terms of the series.

(2)

(Total 5 marks)

Q6.

An arithmetic series has first term a and common difference d .

The sum of the first 31 terms of the series is 310.

(a) Show that $a + 15d = 10$.

(3)

(b) Given also that the 21st term is twice the 16th term, find the value of d .

(3)

(c) The n th term of the series is u_n . Given that $\sum_{n=1}^k u_n = 0$, find the value of k .

(4)

(Total 10 marks)

Q7.

(a) An infinite geometric series has common ratio r .

The first term of the series is 10 and its sum to infinity is 50.

(i) Show that $r = \frac{4}{5}$.

(2)

(ii) Find the second term of the series.

(2)

(b) The first and second terms of the geometric series in part (a) have the same values as the 4th and 8th terms respectively of an arithmetic series.

(i) Find the common difference of the arithmetic series.

(3)

(ii) The n th term of the arithmetic series is u_n . Find the value of $\sum_{n=1}^{40} u_n$.

(4)

(Total 11 marks)

Q8.

The 25th term of an arithmetic series is 38.

The sum of the first 40 terms of the series is 1250.

(a) Show that the common difference of this series is 1.5.

(6)

(b) Find the number of terms in the series which are less than 100.

(3)

(Total 9 marks)

Q9.

The arithmetic series

$$51 + 58 + 65 + 72 + \dots + 1444$$

has 200 terms.

- (a) Write down the common difference of the series.

(1)

- (b) Find the 101st term of the series.

(2)

- (c) Find the sum of **the last** 100 terms of the series.

(2)

(Total 5 marks)

Q10.

An arithmetic series has first term a and common difference d .

The sum of the first 29 terms is 1102.

- (a) Show that $a + 14d = 38$.

(3)

- (b) The sum of the second term and the seventh term is 13.

Find the value of a and the value of d .

(4)

(Total 7 marks)

Q11.

The first term of an arithmetic series is 1. The common difference of the series is 6.

- (a) Find the tenth term of the series.

(2)

- (b) The sum of the first n terms of the series is 7400.

- (i) Show that $3n^2 - 2n - 7400 = 0$.

(3)

- (ii) Find the value of n .

(2)

(Total 7 marks)

Q12.

An arithmetic series has fifth term 46 and twentieth term 181.

- (a) (i) Show that the common difference is 9.

(3)

- (ii) Find the first term.

(1)

- (b) Find the sum of the first 20 terms of the series.

(2)

- (c) The n th term of the series is u_n .

Given that the sum of the first 50 terms of the series is 11525, find the value of

$$\sum_{n=21}^{50} u_n$$

(2)

(Total 8 marks)

Q13.

The n th term of an arithmetic sequence is u_n , where

$$u_n = 90 - 3n$$

- (a) Find the value of u_1 and the value of u_{20} .

(2)

- (b) Write down the common difference of the arithmetic sequence.

(1)

- (c) Given that $\sum_{n=1}^k u_n = 0$, find the value of k .

(3)

(Total 6 marks)